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ABAP Business Application Programming Interface (BAPI)

Deloitte Technology Academy (DTA)

Agenda

Topic	Descriptions	Duration
Business Application Programming Interface (BAPI) Overview	Business Object Repository, What Is BAPI?, Business Object With BAPIs and Associated Function Modules, Benefits of BAPI, Interfaces to R/3 System, Return Structure, Transactions for BAPI, Step by Step Procedure, Define Structures for BAPI and Develop a Function Module, Create Application Programming Interface (API) Method Using BAPI Wizard and BAPI Explorer	4 hours

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Learning Objectives

By the end of this session, you will be able to:

- Describe the business object repository and BAPI
- Discuss the benefits of BAPI
- Describe BAPI structure
- Use BAPI explorer
- Describe how to develop a function module and create an API method using the BAPI wizard

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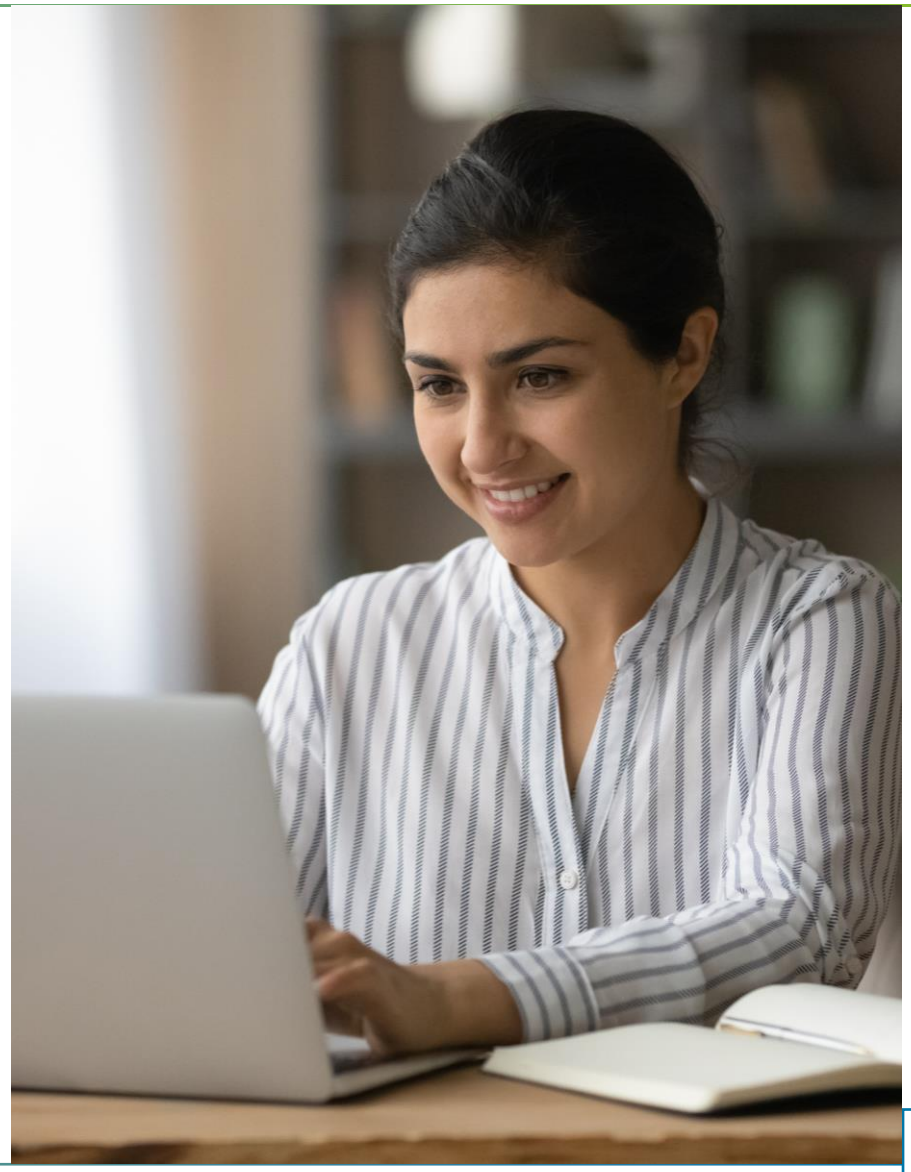
Business Application Programming Interface (BAPI) Overview



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Business Object Repository

- All Systems, Applications, and Products in Data Processing (SAP) business object types, SAP interface types, and their methods are defined and described in the R/3 Business Object Repository (BOR).
- The business object repository was introduced in R/3 release 3.0, at the same time as SAP business objects and SAP business workflow was introduced. Presently, the BOR is mainly used by SAP business workflow.
- The BOR has two essential functions:

It defines and describes SAP business objects, SAP interface types, and their BAPIS.

It creates instances of SAP business objects.

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What Is a BAPI?

- BAPIs are standardized programming interfaces (methods) enabling external applications to access business processes and data in the R/3 system
- BAPIs are defined in the Business Object Repository (BOR) as methods of SAP business objects or SAP interface types and are implemented as function modules.
- The separation of a BAPI definition from its actual implementation enables you to access a BAPI in two ways.
  - You can call the BAPI in the BOR through object-oriented method calls.
  - You can make Remote Function Call (RFC) calls to the function module on which the BAPI is based.
- BAPI can be called within the R/3 system from external application systems and other programs.

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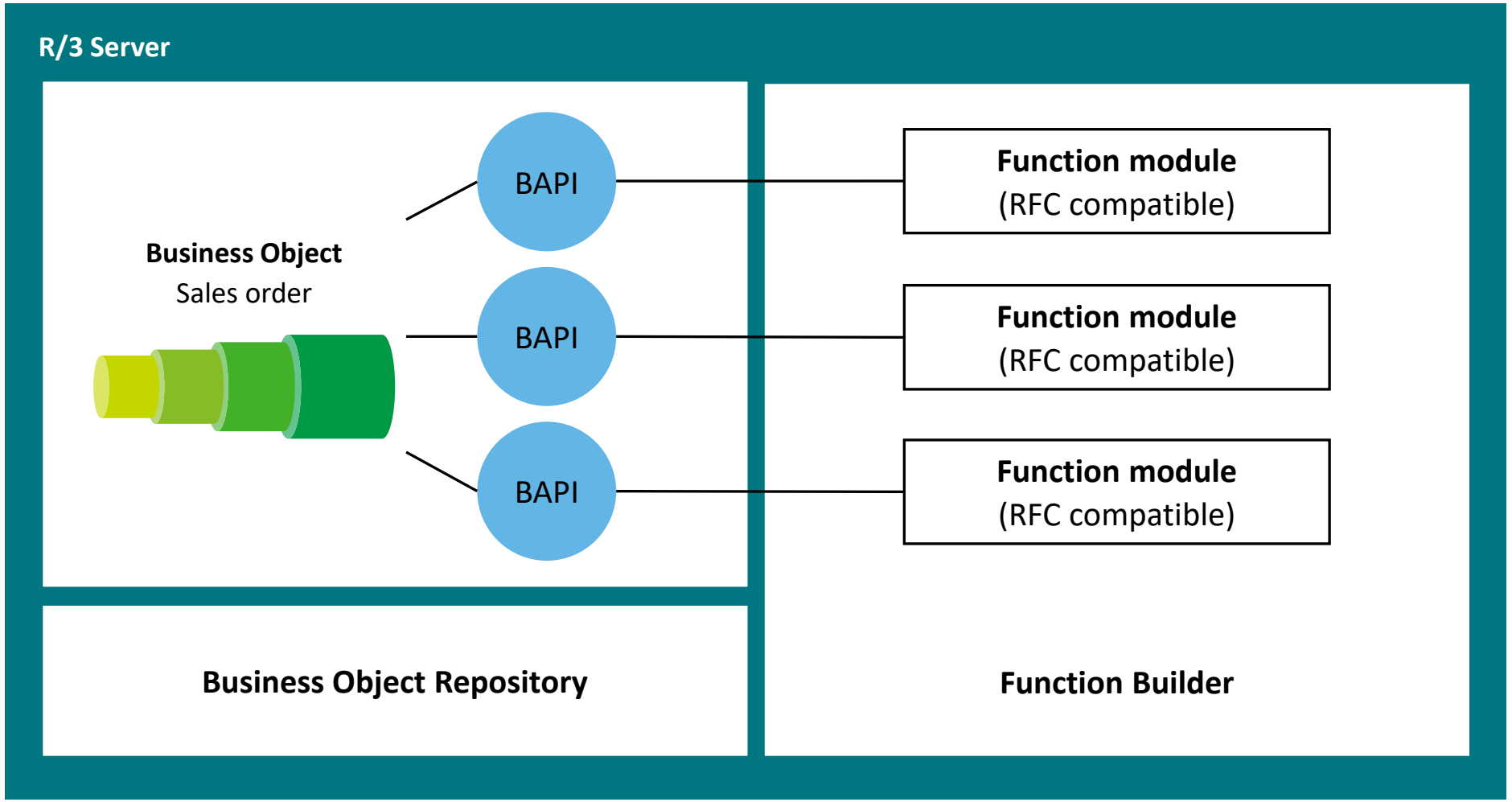
Business Object With BAPIs and Associated Function Modules



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Benefits of BAPI

Can be used in diverse languages or development environments

Helps in reducing development cost

Helps in minimizing maintenance cost

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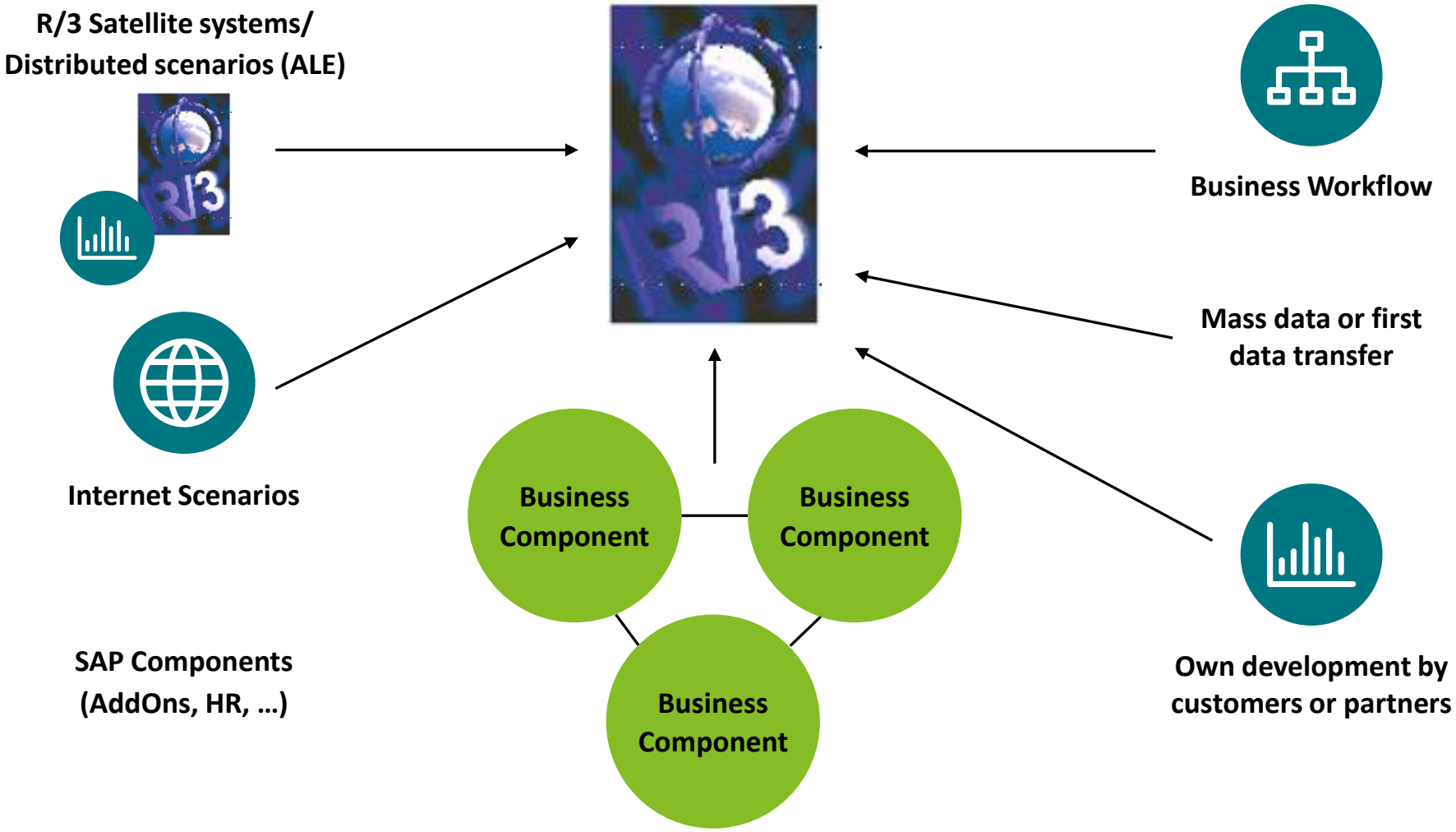
BAPI—Interfaces to R/3 System



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BAPI Return Structure

- Usually, the return structure is a work area or a table sometimes.
- Message types
  - S
  - E
  - W
  - I
  - A

Message	Message Text
Log_no	Application log number
Log_Msg_No	Application log message serial number
Message_V1 – V4	Message variables

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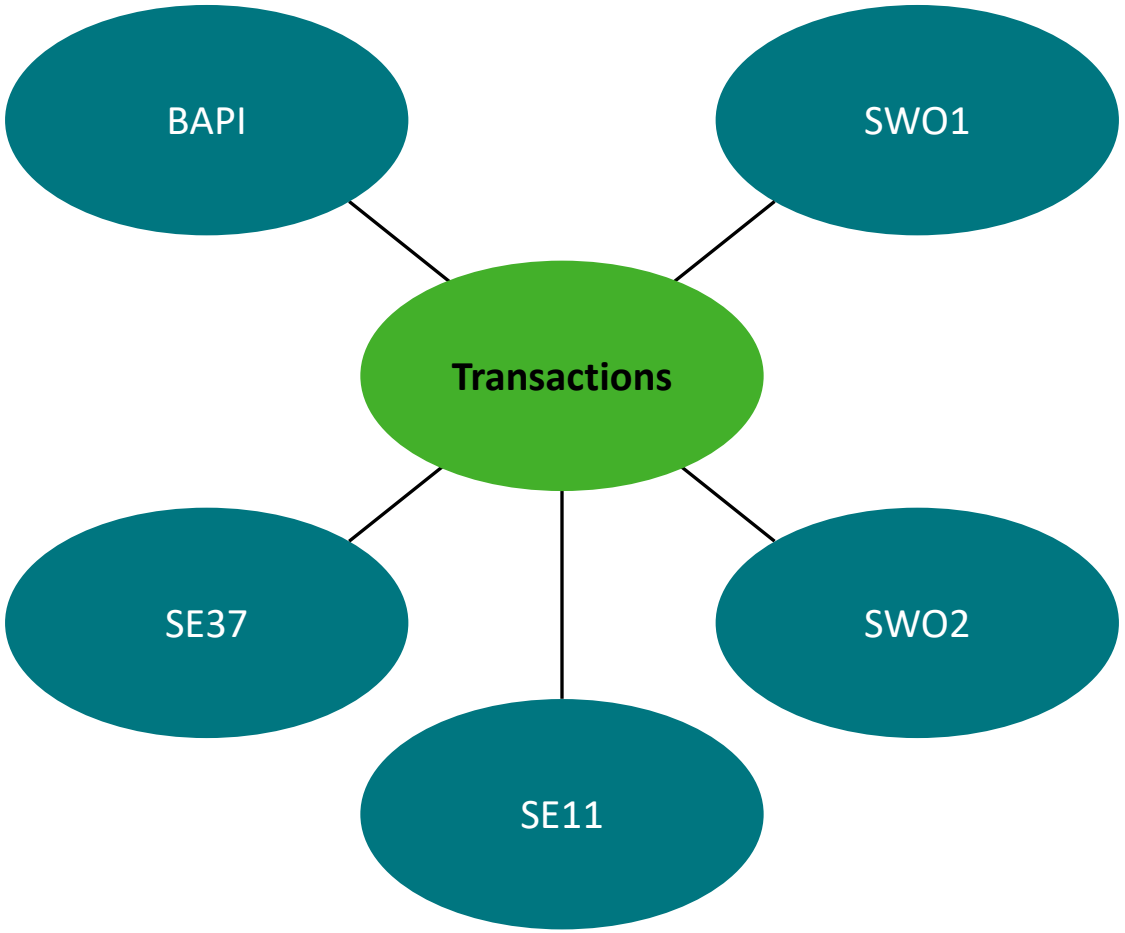
Transactions for BAPI



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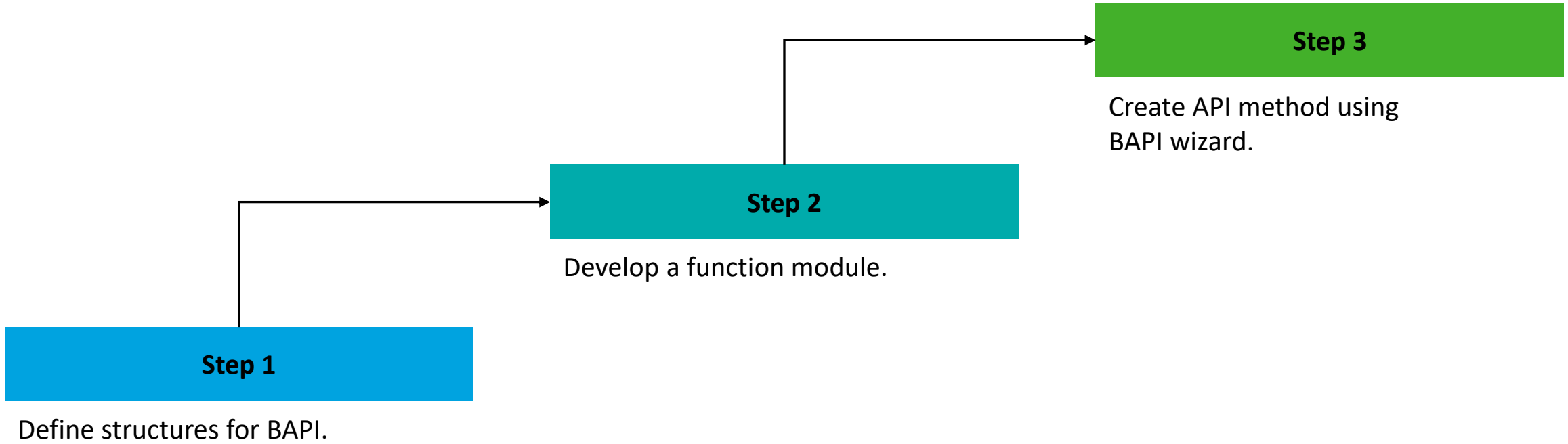
BAPI—Step by Step Procedure



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Define Structures for BAPI

- Define structures and tables for the parameters of the BAPI.
- Use transaction code SE11 to create structures and tables.
- Define a structure for every parameter in the BAPI.
- Do not use the same structures used in existing applications as BAPI structures are frozen when they are released and there are restrictions on changing the structures.

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Develop a Function Module

- Each BAPI must be defined in its own function group.
- Select the processing type as ‘Remote Enabled’ to make the function module a Remote Function Call (RFC).
- Import/export parameters can only be BY VALUE for RFCs.

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Create API Method Using BAPI Wizard

- BAPI wizard is used to exposing the remote function module as a BAPI.
- BAPI wizard will generate additional code to make the function module a valid method of the BOR.
- This allows BAPI to be called by outside programs.

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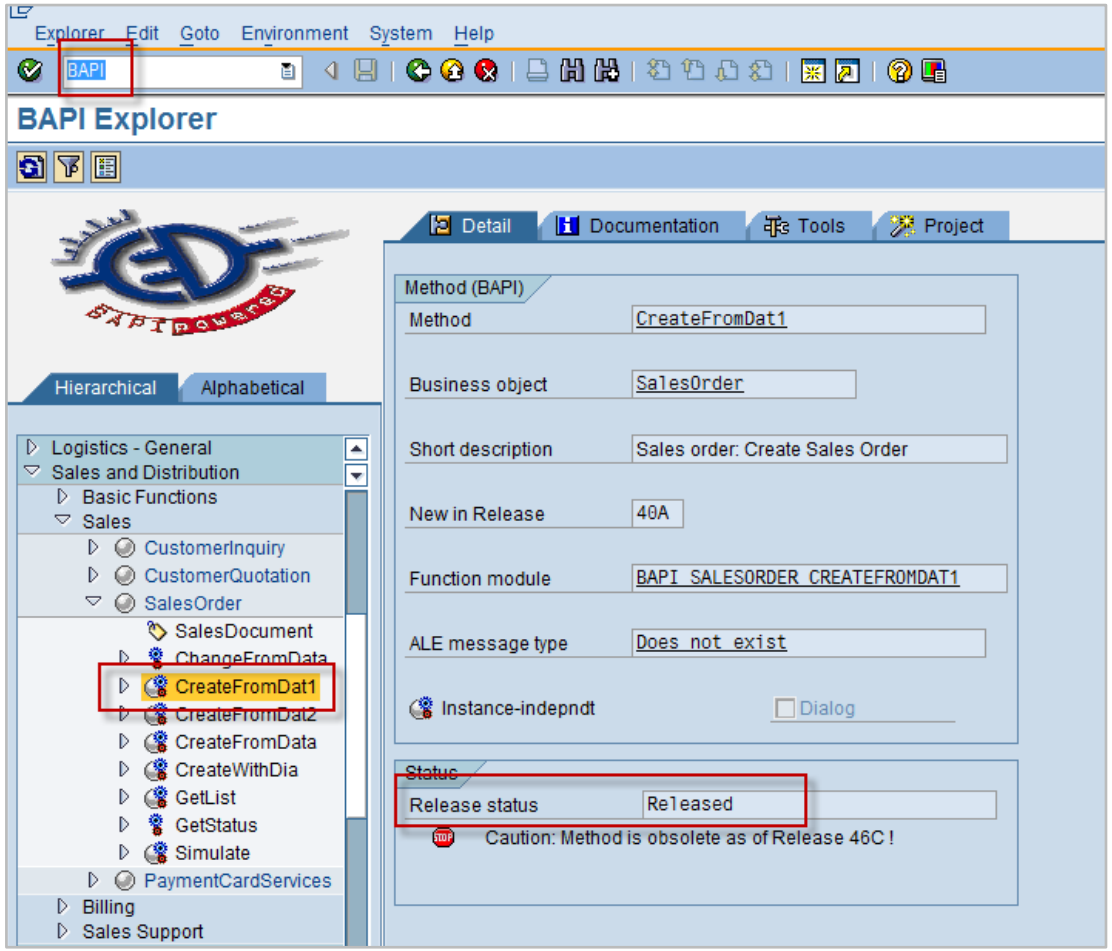
BAPI Explorer

- BAPI explorer is a tool to get all the required information to use any BAPI
- BAPI explorer provided all the tools used to create BAPIs and the required development objects in an integrated programming environment
- Tcode : BAPI



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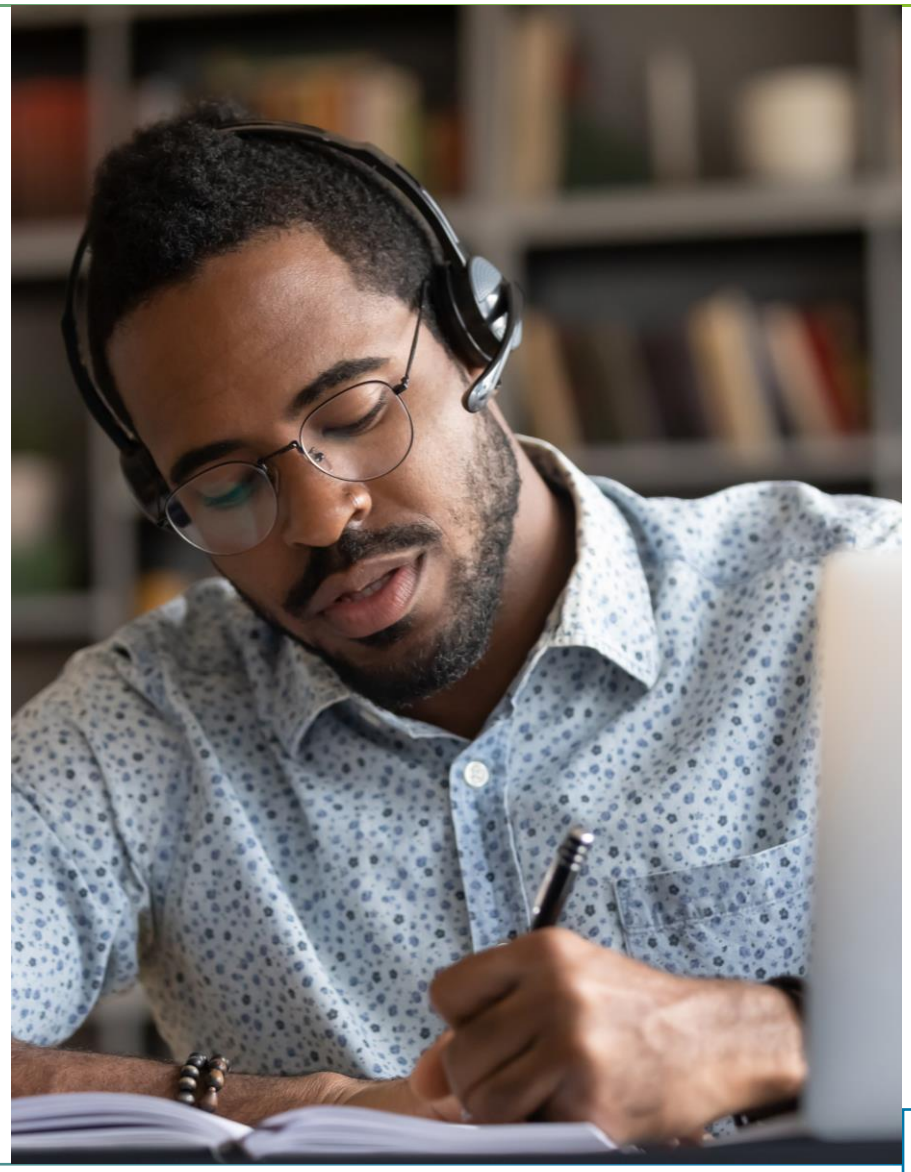
Hands-On Labs



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Hands-On Activity 1

Activity Details	
Problem Statement	Write a program to create a business object "Equipment" using a BAPI call. It should be created with General and Location data. Find out the field lengths using the IE03 transaction. • Program name to be: ZH<XXX>C_BAPI_<desc> – <XXX> — Last two digits of your SAP user id – <desc> — Meaningful Description • Selection screen – Presentation server file path and file name. The business object "Equipment" is an individual, physical object that is to be maintained independently. It can be installed in a technical system or part of a technical system. You can manage all types of devices as pieces of equipment (for example, production utilities, transportation utilities, test equipment, production resources/tools, buildings, PCs).

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Hands-On Activity (Cont.)

Activity Details	
Problem Statement	**Transaction—IE01**: No pre-validations are required. Pass set of Header, item, and schedule line details from the input file to the BAPI. **Header structure and partner function**:

Field name	Description
EQKT-EQKTX	Equipment description
EQUI-EQTYP	Equipment category
EQUI- HERLD	Country of manufacture
ILOA- SWERK	Maintenance plant
ILOA – STORT	Location of maintenance object
ILOA –ABCKZ	ABC indicator

File structure: The file contains the general information and location data of the equipment.

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Hands-On Activity (Cont.)

Activity Details	
Problem Statement	**Error handling**: Display the records which have errors after BAPI execution as a list output with an Error description.
Output Display	* Total number of equipment created successfully: <No of equipment created> * Equipment numbers created: <EQUNR1> < EQUNR2> * Error records: <Print all records which are errors in the BAPI: Record with all fields from the file with the Error message

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Summary

Here are the key learning points of the module.

- Business object repository defines SAP business objects, SAP interface types, and their BAPIs.
- BAPI is defined in the Business Object Repository (BOR) as methods of SAP business objects or SAP interface types and is implemented as function modules.
- BAPI can be used in diverse languages or development environments. Each BAPI must be defined in its own function group.
- BAPI wizard is used to exposing the remote function module as a BAPI.

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